

Candidate Seat No: _____

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor of Science in Information Technology (B.Sc. IT)

Semester-II University Examination May-2018

CA 406 Operating System Concepts

Date: 07.05.2018, Monday

Time: 10.00 a.m. to 01.00 p.m.

Maximum Marks: 70

Instructions:

1. Attempt both sections in separate answer books.
2. The figure to the right indicates marks.
3. Make a suitable assumption when necessary.

SECTION-I

Q.1 [A] Column A lists of OS concepts and B with the characteristics of them. [05]
Match "A" with "B".

A

- (1) Serial Processing
- (2) Time Sharing
- (3) Batch OS
- (4) Multiprogramming
- (5) OS Kernel

B

- (a) CPU Scheduling
- (b) Maximum Processor Utilization
- (c) No Operating Systems.
- (d) Reliability issue.
- (e) Uniprogramming

[B] Fill in the blanks with appropriate answer. [06]

1. _____ provides an interface to the services made available by an operating system.
2. _____ program switches the processor from one process to another.
3. An _____ is the memory address of an interrupt handler.
4. _____ is the listing of a sequence of instructions that execute for a process.
5. In Round Robin algorithm _____ interrupt generates.
6. The decision mode of HRRN policy is _____.

Q.2 [A] Explain five state process model with state transition diagram, states, [06]
transitions and limitations.

[B] Give the answers of the questions based on HRRN and SPN scheduling [06]
policies in the following example.

Processes	Arrival Time	Service Time
A	0	6
B	3	2
C	4	4
D	6	1
E	7	6

- (a) What is the waiting time of Process E?
- (b) What is the normalized turn round time of Process C?
- (c) What is the finish time of Process D?

OR

Q.2[A] What do you mean by Operating Systems? List out the objectives of it and [06]
explain the role of it in (a) Program Execution and (b) I/O access.

[B] Explain any twelve reasons of process termination. [06]

for all files called as _____.

- a) bin
- c) home

- b) dev
- d) root

Q.5 [A] What is called memory management? Explain memory management requirements. **[06]**

[B] Explain in detail all the features of UNIX operating system. **[06]**

OR

Q.5 [A] Explain following commands with options and suitable examples: **[06]**

- a) ls
- b) grep
- c) sort

[B] Explain UNIX directory structure with diagram. **[06]**

Q.6 Answer following questions in detail (Any three) [12]

1. What is called fragmentation? Explain its types.
 2. Give difference between Logical address and Physical address
 3. Write a shell script to find largest among three numbers entered by user.
 4. Explain with diagram UNIX architecture.
 5. What is vi editor? Explain in detail modes of vi editor.
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